Diabetes Insights – An Interactive Dashboard for Public Health Analytics

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1 Abstract

The Diabetes Insights project presents an interactive data visualization dashboard designed to explore diabetes prevalence and associated risk factors across different population groups. Leveraging publicly available health data, the dashboard enables users to investigate how variables such as age, gender, education, BMI, physical activity, smoking status, and healthcare access relate to diabetes outcomes.

Built using HTML, CSS, JavaScript (with D3.js and Chart.js), the dashboard supports intuitive, responsive, and user-driven exploration through interactive charts, filtering, and detail-ondemand features. Optional backend support via Flask and Python allows for dynamic data processing and API integration when needed.

This tool is intended to serve a wide range of users including healthcare professionals, researchers, policymakers, and the general public by offering a clear and engaging platform to understand population health patterns and disparities. Through thoughtful design grounded in information visualization principles, the Diabetes Insights dashboard transforms complex health data into actionable insights that can inform awareness, research, and targeted intervention strategies.

2 Introduction

Diabetes is one of the most prevalent chronic conditions worldwide, affecting over 500 million individuals and posing a significant burden on public health systems. Early diagnosis, prevention, and management of diabetes require a deep understanding of its underlying risk factors, including not only clinical indicators but also behavioral and social determinants such as physical activity, access to care, and socioeconomic status.

Despite the abundance of health data available today, most diabetes dashboards and analytic tools remain limited in scope. Many platforms offer static views, predefined charts, or clinical data with little interactivity. They often overlook key lifestyle and social variables and fail to support multi-dimensional exploration or realtime comparison. These limitations create a gap between raw data and meaningful, actionable insight especially for non-technical users like community health workers or patients.

To address these challenges, we developed Diabetes Insights an interactive, web-based dashboard designed to support rich, userdriven exploration of diabetes-related data. Our system enables filtering by demographics (age, gender, education), behavioral factors (BMI, smoking, physical activity), and healthcare accessibility indicators. Through dynamic visualizations such as scatterplots, heatmaps, line charts, and donut charts, the dashboard facilitates data-driven storytelling across different user groups.

By combining clear visual encoding with interactive features such as brushing, hover tooltips, and real-time filtering, our dashboard aims to bridge the usability gap in traditional health tools. This paper presents the design rationale, implementation methodology, evaluation results, and future roadmap of the Diabetes Insights platform.

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3 Related Work

A major driving force behind the development of our new diabetes insights dashboard came from a careful review of existing work spanning research dashboards and publicly available tools. Through this review, we noticed several recurring problems that shaped the design goals for our project. First, many dashboards, like those created by Martinez et al. (2018) [1] and Koopman et al. (2011) [2], mainly offered static visualizations. Users could only view predefined charts and had little to no ability to interact with the data, filter it, or ask deeper questions on their own. Second, we found that most systems only allowed for single-variable analysis. Dashboards such as the OpenMRS Diabetes Dashboard [4] and the Danish Diabetestal Dashboard [8] typically showed relationships based on just one factor at a time like age or blood pressure without letting users explore how multiple factors interact to influence diabetes risk. Another major gap was the lack of behavioral and social context. Critical risk factors like income level, education, smoking habits, number of unhealthy physical health days, and healthcare access were often missing. This was especially clear in tools reviewed by Jørgensen et al. (2019) [5]and open projects like Nightscout [11], where the focus stayed mostly on clinical or biological measures. We also noticed that many dashboards were designed for a very specific audience. Some targeted only patients (like Martinez et al.) [1] while others catered solely to clinicians (like Koopman et al.) [2]. Very few dashboards tried to make their insights accessible to researchers, policymakers, students, or the general public. When it came to exploration capabilities, most tools were still limited. Only a few advanced systems like the MOSAIC Project [7] and Directive Explanations Dashboard (Bhattacharya et al., 2023) [10] allowed users to interact with the data in a meaningful way through features like real-time filtering or what-if analysis. But even then, users were often left as passive viewers instead of active explorers. In some cases, the problem was the opposite overly complex dashboards. Predictive analytics platforms like MOSAIC [7] and Directive Explanations [10] were difficult to use without technical expertise, making them less suitable for broader, non-specialist audiences. Another important missing piece was the absence of health inequity visualization. Dashboards like 1120-P Diabetestal [8] and the GDM Dashboard [9] mostly focused on clinical statistics without showing how social factors like education, income, or gender disparities impact diabetes outcomes. Altogether, these limitations highlighted the need for a modern, interactive, exploratory dashboard one that integrates medical, behavioral, and social data in a way that's accessible to researchers, healthcare providers, policymakers, and even the general public. That gap is exactly what we set out to address with our new system.These findings informed our dashboard design goals.

4 Implementation

The Diabetes Insights dashboard was developed as a full-stack web application with a clean, modular architecture supporting dynamic, realtime data visualization. The system integrates a Python-based backend with a responsive JavaScript frontend, ensuring smooth interaction and rapid updates based on user inputs.

4.1 Technology Stack

The implementation utilizes the following technologies:

- Frontend: HTML, CSS, JavaScript, D3.js, Chart.js
- Backend: Python, Flask, Flask-CORS
- Data Handling: Pandas (for preprocessing), JSON (for data exchange)
- Testing Tools: Postman (API testing), browser dev tools
- Version Control: Git and GitHub

4.2 System Architecture

The dashboard follows a client-server model:

- The backend (Flask) serves RESTful API endpoints that process, filter, and return JSON-formatted data subsets based on user filters (e.g gender, age, education).
- The frontend dynamically renders visualizations using D3.js and Chart.js based on responses received from the backend.
- Filtering controls and dropdowns in the frontend trigger fetch requests to the API, ensuring real-time updates without page reloads.

4.3 Dashboard Structure

The system is segmented into four key pages:

1. Overview Page: Displays total patient count, high-level trends, and navigation cards.
2. Diabetes Trends Page: Interactive line chart showing prevalence by age, with filters for gender and education.
3. Health-Related Factors Page: Scatterplot for BMI vs. physically unhealthy days, and donut charts showing behavioral and access indicators.
4. Health Disparities Page: Grouped and stacked bar charts revealing diabetes trends across income, education, and gender.

4.4 Data Preprocessing

The BRFSS diabetes dataset was cleaned using Pandas to remove duplicates and handle missing values. Variables were recoded into readable labels (e.g education levels, income groups) for intuitive mapping in visualizations. The cleaned dataset was stored in a CSV file and loaded into the backend for API-based access.

4.5 Interactive Features

Key interactive components implemented:

- Brushing and zooming in scatterplots for detailed exploration
- Hover tooltips showing contextual data (e.g diabetes % in age group)
- Toggle switches to change between percentage and count views
- Real-time filtering controls across pages

4.6 Deployment

The application is hosted locally for demonstration and testing. All dependencies are managed via Python virtual environments and npmbased static scripts. GitHub hosts the public code repository, with deployment instructions included in the README.

Repository: https://srihari0520.github.io/assets/projects/diabetes-insights

https://srihari0520.github.io/assets/projects/diabetesinsights

5 Methodology

The development of the Diabetes Insights dashboard followed a usercentered and task-driven design approach, aimed at addressing the limitations of existing health dashboards and making complex data more accessible and actionable.

5.1 User Goals Analysis

The intended users of the dashboard include healthcare professionals, public health researchers, policymakers, and the general public. Through informal interviews and peer feedback sessions, we identified the following key user needs:

- Understand diabetes prevalence across demographics such as age, gender, and education.
- Analyze behavioral risk factors like smoking, BMI, and physical inactivity.
- Compare groups and explore disparities based on income, education, and gender.
- Interactively filter, browse, and examine subsets of data without requiring programming skills.

5.2 Heuristic Evaluation

We conducted a heuristic evaluation based on Nielsen's usability principles. This revealed several gaps in conventional dashboards: limited filtering options, static visuals, poor responsiveness, and missing detailon-demand. These findings guided the interactive and modular design of our dashboard.

5.3 Task and Visualization Mapping

Each visualization was mapped to a specific user task and encoded using effective visual variables (e.g color, length, position). The table below summarizes this mapping:

Table 1: Mapping of User Tasks to Visual Components

User Task	Chart Type	Encoding	Interaction
Explore diabetes by age	Line Chart	X: Age, Y: Prevalence (%)	Filters: Gender, Education
Compare behavioral risk	Scatterplot	X: BMI, Y: Phys. Health Days, Color: Diabetes Type	Brushing, Tooltip
Understand gender disparity	Stacked Bar Chart	X: Education, Y: %, Fill: Gender	Hover, Filter
Analyze income effect	Grouped Bar Chart	X: Income, Y: Prevalence, Color: Diabetes Type	Dropdown Filter
Assess access to care	Donut Chart	Segment: Status (Yes/No)	Hover Tooltip

5.4 Thematic Segmentation

To avoid cognitive overload and enable deep exploration, we structured the dashboard into four thematic pages: Overview, Diabetes Trends, Health-Related Factors, and Health Disparities. Each page is tailored to a focused analytical goal and offers relevant filters and visual elements, improving user experience through contextual separation.

5.5 Visual Design Considerations

The dashboard adopts a consistent color palette, clear legends, and intuitive layout. Visual grouping and proximity (following Gestalt principles) were used to link related filters and views. Animated transitions and hover-based detail-on-demand were added to promote fluid exploration.

6 TESTING

The Diabetes Insights dashboard was tested iteratively throughout development to ensure functionality, usability, and responsiveness across various devices and user scenarios. Both technical and usercentered testing approaches were applied.

6.1 Functional Testing

We conducted component-wise testing of the dashboard to verify:

- API correctness: All backend endpoints were tested using Postman to ensure accurate data retrieval, filtering, and JSON formatting.
- Chart rendering: All visualizations (line, bar, scatter, heatmap, donut) were tested for correct data binding and responsiveness to filter inputs.
- Filter control functionality: Age, gender, and education dropdowns were tested to ensure they triggered real-time updates to visuals without page reloads.
- Cross-page navigation and layout consistency: Each thematic page was verified for correct routing and layout rendering.

6.2 Usability Testing

We conducted informal usability testing sessions with 6 peer users (students and non-technical users). Participants were asked to perform tasks such as:

- Identify the most affected age group.
- Compare diabetes rates between genders by education level.
- Explore the correlation between BMI and physical health days.
- Filter by education and income level to observe disparities.

Feedback from these sessions revealed usability enhancements, including:

- Enlarging chart legends and axis labels for readability.
- Improving color contrast in heatmaps.
- Clarifying default states when no filters are selected.

6.3 Cross-Device Testing

The dashboard was tested on multiple devices and screen sizes including:

- Desktop browsers (Chrome, Firefox, Edge)
- Tablets (iPad)
- Mobile (iPhone, Android)

Responsive layout was verified using browser developer tools and real devices. Minor adjustments to padding and card layouts were made to improve mobile usability.

6.4 Performance Testing

Chart load times were measured using browser dev tools. The backend

API response time remained under 300ms on average for all filter combinations. Visual rendering and transitions remained smooth under typical dataset load.

7 RESULTS AND DISCUSSION

The Diabetes Insights dashboard successfully met its design goals by enabling users to explore diabetes-related patterns interactively and intuitively. It transformed a static health dataset into a dynamic visual platform supporting analysis of population-level disparities, behavioral risk factors, and demographic trends.

7.1 Insights from the Dashboard

Several key patterns were identified through dashboard use:

- Age and Diabetes: Line charts revealed that diabetes prevalence increases sharply after age 45, with the highest risk observed in individuals aged 65 and older.
- Gender and Education Disparity: Stacked bar charts showed that females with lower education levels tend to have higher diabetes rates compared to their male counterparts, highlighting a gender-based education gap.
- Behavioral Risk Analysis: The BMI vs. physical health scatterplot indicated a visible cluster of diabetic individuals in the highBMI and high unhealthy days quadrant, reinforcing known behavioral risk associations.
- Income and Access Disparities: Grouped bar and heatmap visualizations demonstrated that lower-income groups reported higher diabetes prevalence, along with greater difficulty accessing healthcare due to cost or availability.

7.2 Design Successes

The thematic page structure proved effective in guiding users through progressive exploration:

- Overview cards provided a quick orientation to major indicators.
- Each chart was tailored to its analytical purpose, using appropriate encodings (e.g color for diabetes status, height for proportion).
- Filtering tools, toggles, and interactive elements empowered users to investigate questions independently without needing technical expertise.

7.3 Limitations and Challenges

While the dashboard achieved its core objectives, several limitations remain:

- Single-year dataset: The dashboard currently operates on data from a single year (2015). Including multi-year data could reveal time-based trends.
- No predictive analytics: The dashboard focuses on exploratory analysis; predictive modeling (e.g diabetes risk scoring) is not yet included.
- Mobile optimization: Although responsive, certain charts (e.g heatmaps and clustered bar charts) are harder to read on smaller screens.
- Data granularity: Some subgroups (e.g elderly females with no education) had limited representation, affecting statistical reliability.

7.4 User Feedback

Usability testing and peer feedback confirmed that:

- Filters and brushing significantly enhanced data exploration.
- The page-wise layout reduced cognitive load and improved focus.
- Non-technical users appreciated the simplicity of interactions and consistent color encodings.

Overall, the dashboard demonstrates how interactive visual analytics can bridge gaps in public health data accessibility, enabling more equitable insight generation and informed decision-making.

8 Conclusion

The Diabetes Insights dashboard demonstrates how interactive visual analytics can significantly enhance the exploration and understanding of complex public health data. By integrating behavioral, demographic, and socioeconomic dimensions, the dashboard offers a comprehensive view of diabetes risk factors that goes beyond traditional static reports. Throughout the development process, we applied user-centered design principles, ensuring that each visualization and interface component was aligned with specific user tasks. The modular, thematic structure enabled focused analysis, while interactive features like brushing, real-time filtering, and hover-based detail-on-demand empowered users to engage with the data more effectively.

Our implementation spanning full-stack technologies including Flask, D3.js, and Chart.js ensures responsiveness, accuracy, and extensibility. Usability testing confirmed the effectiveness of the dashboard for both technical and non-technical users, including healthcare professionals, researchers, and the general public.

While the current system is limited to exploratory analysis on singleyear data, it provides a solid foundation for future work. Potential next steps include incorporating multi-year datasets, enabling timeseries comparisons, and integrating machine learning models for predictive diabetes risk scoring.

Overall, the dashboard successfully bridges gaps in accessibility and interactivity in diabetes data analysis, supporting data-driven decisionmaking for public health and clinical research.

9 Future work

- Voice or Chat-Based Navigation

  To improve accessibility and user experience, we will add support for voice commands or chatbot style navigation. This would allow users especially those with visual impairments or limited tech skills to explore data using natural language, making the dashboard more inclusive and intuitive.

- Multilingual Support

  Adding translation support will enable non-English speaking users to interact with the dashboard in their native language. This feature is especially important for public health tools aimed at diverse communities and would greatly increase the dashboard's global usability.

- User-Driven What-If Analysis

  This feature would let users modify variables (e.g reduce BMI or increase physical activity) and instantly see how those changes affect diabetes risk. It supports scenario testing and encourages interactive learning, helping users understand the impact of lifestyle choices on health outcomes.

10 Appendix

10.1 A. Dashboard Screenshots

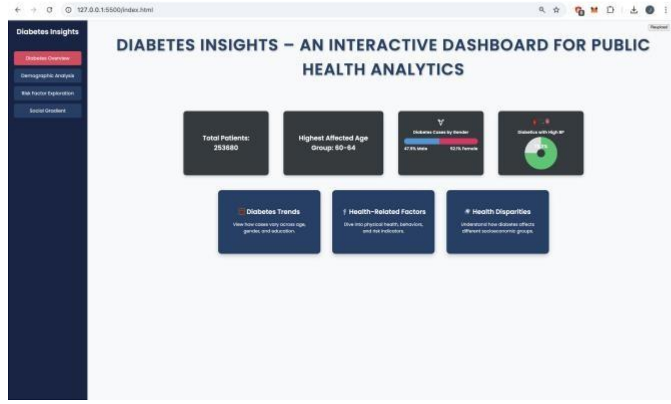

Fig. 1: Overview Page showing total patients, most affected groups, and navigation to deeper analysis.

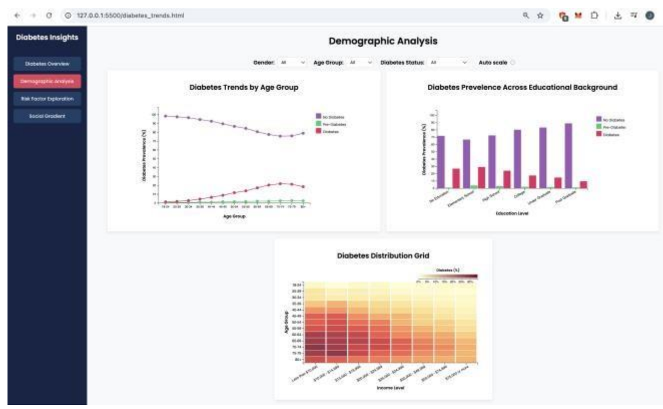

Fig. 2: Diabetes Trends Page – Multi-line chart with filters for age, gender, and education level.

Fig. 3: Health-Related Factors Page – Scatterplot and donut charts exploring behavioral risk indicators.

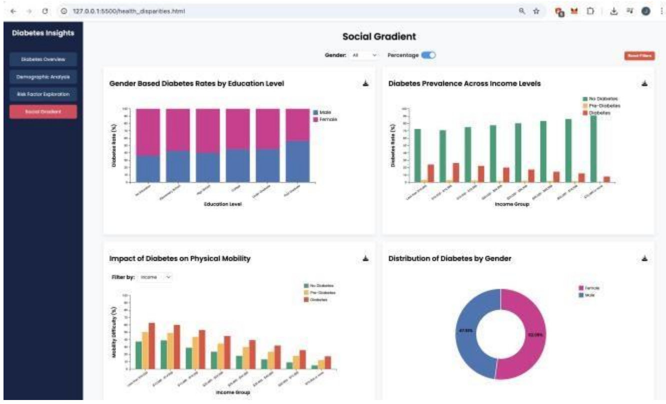

Fig. 4: Health Disparities Page – Gender and income-based comparisons with grouped/stacked bars.

10.2 B. Gantt Chart – Project Timeline

Fig. 5: Gantt Chart outlining key project milestones: setup, preprocessing, design, implementation, testing, and documentation.

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